MOHAMMED IRFAN KURPATH

PhD Student, Computer Vision · MBZUAI, Abu Dhabi

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PROFILE

I'm a PhD student in Computer Vision at MBZUAI in Abu Dhabi. My research centers on building multimodal AI systems for **proactive, personalized assistance**: systems that can see, listen, remember, reason, and speak.

Interests: omni-modal LLMs, agents and tool use, omni-modal embedding and retrieval, long-term multimodal memory, multimodal reasoning benchmarks, real-time speech-to-speech, multilingual MLLMs, healthcare and assistive AI.

PUBLICATIONS

1. M. I. Kurpath*, J. M. Kaithakkodan*, J. Zhou, S. S. Mullappilly, M. Almansoori, N. Ahsan, B. Kalmakhanbet, S. Shikhar, R. Lalla, J. Lahoud, M. Awad, F. S. Khan, S. Khan, R. M. Anwer, H. Cholakkal. **LongShOT: A Benchmark for Omni-Modal Reasoning in Long Videos.** **NeurIPS 2026** (to appear). [Project Page]
2. M. I. Kurpath, J. M. Kaithakkodan, S. S. Mullappilly, I. Laptev, H. Cholakkal. **Omni-Embed-Mini: Binding Modalities Without Forgetting via Dense Distillation.** **EMNLP 2026 Findings** (to appear). [Project Page]
3. S. S. Mullappilly*, M. I. Kurpath*, J. M. Kaithakkodan, O. Mohamed, M. Zidan, F. Khan, S. Khan, R. Anwer, H. Cholakkal. **MediX-R1: Open Ended Medical Reinforcement Learning.** **EMNLP 2026 Findings** (to appear). [Project Page]
   8B model outperforms the 27B MedGemma; 2B variant runs on-device on a phone.
4. H. Rasheed*, M. I. Kurpath*, B. Ren, H. Cholakkal, F. S. Khan, S. Khan. **Hard Vision, Easy Vision: What GPT-6 Astra Reveals Across Computer Vision.** **arXiv preprint**, 2026. [Project Page]
5. S. S. Mullappilly*, M. I. Kurpath*, S. Pieri, S. Y. Alseiari, S. Cholakkal, K. M. Aldahmani, F. S. Khan, R. M. Anwer, S. Khan, T. Baldwin, H. Cholakkal. **BiMediX2: Bio-Medical EXpert LMM for Diverse Medical Modalities.** **EMNLP 2025 Findings**. [Code]
6. S. Shikhar, M. I. Kurpath, S. S. Mullappilly, J. Lahoud, F. S. Khan, R. M. Anwer, S. Khan, H. Cholakkal. **LLMVoX: Autoregressive Streaming Text-to-Speech Model for Any LLM.** **ACL 2025 Findings**. [Project Page]
7. Y. Kryklyvets, M. I. Kurpath, S. S. Mullappilly, J. Zhou, F. S. Khan, R. M. Anwer, S. Khan, H. Cholakkal. **MAviS: A Multimodal Conversational Assistant For Avian Species.** **EMNLP 2025**. [Code]
8. B. S. Shafique, A. Vayani, M. Maaz, H. A. Rasheed, D. Dissanayake, M. I. Kurpath, et al., S. Khan, F. S. Khan. **A Culturally-Diverse Multilingual Multimodal Video Benchmark & Model.** **EMNLP 2025**. [Project Page]
9. M. I. Kurpath, P. K. Adwai, J. Bodireddy, C. K, N. S. K. **An IMUs and Potentiometer-Based Controller for Robotic Arm-Hand Teleoperation.** **Sensors and Actuators A: Physical**, 367:115019, 2024. [DOI]

* Equal contribution / joint first authorship.

EXPERIENCE

Aug 2025 to Present — **PhD Student, Computer Vision** · MBZUAI, Abu Dhabi (advised by Prof. Hisham Cholakkal)
Focus: multimodal AI systems for personalized assistance: omni-modal embeddings for memory and retrieval, evidence-seeking agents for long-video understanding, and reliable multimodal models for healthcare.
Responsibilities: lead projects end to end, from problem formulation and benchmark design to model training on multi-node GPU clusters, evaluation and paper writing; mentor students and review for CVPR and NeurIPS.

Mar 2024 to Aug 2025 — **Research Associate** · MBZUAI, Abu Dhabi
Focus: medical and speech-enabled multimodal LLMs, and multilingual multimodal evaluation.
Responsibilities: co-led a bilingual Arabic-English medical LMM, contributed to streaming speech generation for LLMs and to the evaluation toolkit of a multilingual video benchmark; built and presented live demos at GITEX, UNGA79, Abu Dhabi Sustainability Week and Machines Can Think; mentored an MSc thesis and UGRIP interns.

Jul 2022 to Feb 2024 — **Project Associate** · CEET, IIT Madras Research Park, Chennai
Focus: multilingual speech and sign-language technology for Indian languages.
Responsibilities: built voice-command banking across seven Indian languages (ASR, MT, TTS and transliteration), Indian Sign Language recognition from 3D pose, and a speech-controlled wheelchair guide with a real-time chatbot; mentored summer interns.

AWARDS & GRANTS

2026 **Wearable AI Grand Challenge @ ECCV 2026**: **1st place**, EgoConv track (StreamShot: streaming multi-turn conversation over long egocentric video) and **runner-up**, EgoLongQA track (LongShotV2: long egocentric video QA). Entries build on LongShOT (first author, NeurIPS 2026).

2025 **NVIDIA Academic Grant**: to advance clinical reasoning capabilities of healthcare LLMs and LMMs. PI: Dr. Hisham Cholakkal. Role: co-lead student.

2025 **Meta Regional Research Grant**: for Project OMER, building multimodal LLMs for smart wearables. PI: Dr. Hisham Cholakkal. Role: lead student.

2024 **Meta Llama Impact Innovation Award**: for the project BiMediX2 (EMNLP 2025 Findings), building an Arabic-English bilingual medical LMM. PI: Dr. Hisham Cholakkal. Role: co-lead student.

DEMOS & EXHIBITIONS

- **Personalized wearable assistants for elderly care** with smartwatches, smart glasses, and non-invasive EEG devices, at **GITEX 2025** and **Machines Can Think (MCT) 2026**.
- **Multilingual multimodal LLM integrated with a robotic dog** (Unitree GO2) that holds **interactive conversations** and **carries out requested physical actions** while conversing, at **GITEX 2024** and **Abu Dhabi Sustainability Week 2025**.
- **BiMediX2 assistant** deployed over **Telegram**, presented at the **AI4SD event at UNGA79** and **GITEX 2024**. [Demo]

EDUCATION

2025 to Present **PhD, Computer Vision**, MBZUAI — GPA: 3.77/4

2017 to 2021 **B.Tech, Electrical and Electronics Engineering**, NIT Puducherry — CGPA: 8.66/10

SERVICE & MENTORING

- **Reviewer**: CVPR, NeurIPS, and Springer Nature journals.
- **Mentor**, **AI for Africa Bootcamp 2026** (MBZUAI × African Leadership Academy, Johannesburg): graduate mentor to high school students.
- **Mentor**, MBZUAI: MSc thesis (MAviS, EMNLP 2025) and UGRIP undergraduate interns.
- **Mentor**, CEET-IIT Madras Research Park: summer interns.

SKILLS

Languages & Tools Python, Git, Linux
Deep Learning PyTorch, Hugging Face Transformers, OpenCV, vLLM
Training & Infra LLM and multimodal LLM training and fine-tuning, distributed and multi-node training, SLURM, AWS HyperPod-class GPU clusters, NVIDIA and AMD clusters
Speech ASR, TTS, real-time speech-to-speech pipelines
Robotics & Edge ROS, Unitree GO2 quadruped, NVIDIA Jetson, Raspberry Pi, Arduino, ESP32, embedded sensors, IoT

OTHER PROJECTS

Oct 2024 **Real-time Multimodal Conversational Model**: speech-in, speech-out over a lightweight VLM, integrated into the **Unitree GO2 robot dog** and later extended into the **personalized wearable assistants for elderly care** demonstrated at GITEX 2025 and MCT 2026. [GITEX]

Dec 2022 **Fun With Image Processing (Build Lab Course)**: an introductory, hands-on OpenCV course for beginners from diverse backgrounds, offered by the Build Club at IITMRP with partner colleges. [GitHub]

Mar 2022 **Real-time Expression Recognition**: a CNN for seven facial expressions trained on a curated dataset of 9,460 images, reaching 88.35% validation and 82.58% test accuracy, deployed on webcam and video. [GitHub] [Dataset]